

Muscle Activation in Specific Regions of the Trapezius During Modified Kendall Manual Muscle Tests

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Context: Manual muscle tests (MMTs) are often used when assessing shoulder injuries. For the trapezius, individual MMTs are used to selectively test the upper trapezius region (UTR), middle trapezius region (MTR), and lower trapezius region (LTR). The MMTs for each region are assumed to preferentially recruit the corresponding muscle fibers and produce a maximal contraction; however, whether this is true is unknown.

Objective: To determine if maximal voluntary isometric contractions (MVICs) for the upper trapezius (UT-MVIC), middle trapezius (MT-MVIC), and lower trapezius (LT-MVIC), adapted from the Kendall MMTs, recruited the corresponding trapezius regions.

Design: Crossover study.

Setting: Laboratory.

Patients or Other Participants: A total of young, healthy individuals (10 men, 9 women, 1 not listed; age = 23.9 ± 1.7 years, height = 171.4 ± 9.6 cm, mass = 75.7 ± 11.6 kg).

Intervention(s): Participants performed 3 repetitions of each MVIC. High-density surface electromyography measurements were collected from the UTR, MTR, and LTR.

Main Outcome Measure(s): Root mean square (excitation) of the UTR, MTR, and LTR.

Results: We observed an increase in UTR excitation during the LT-MVIC compared with the UT-MVIC ($P = .016$) and MT-MVIC ($P < .001$). The MTR excitation increased during the MT-MVIC ($P = .001$) and the LT-MVIC ($P < .001$) compared with the UT-MVIC. We also noted an increase in MTR excitation during the LT-MVIC compared with the MT-MVIC ($P < .001$). The LTR excitation increased during the MT-MVIC and LT-MVIC (P values $< .001$) compared with the UT-MVIC.

Conclusions: The UT-MVIC and MT-MVIC did not necessarily recruit the corresponding trapezius regions more than the other MVICs did. Rather, the LT-MVIC appeared to produce the greatest excitation of all trapezius regions. Additional research is needed; however, clinicians should be aware that maximal contractions may not always recruit the desired muscle region.

Key Words: high-density surface electromyography, musculoskeletal assessment, muscle recruitment, back assessment, shoulder assessment, neck assessment

Key Points

- A maximal voluntary isometric contraction (MVIC) of the upper trapezius involving scapular elevation did not produce the greatest excitation of the upper trapezius compared with middle and lower trapezius MVICs.
- An MVIC for the middle trapezius involving resisted extension with the shoulder in 90° of abduction did not produce the greatest excitation of the middle trapezius.
- An MVIC for the lower trapezius involving resisted extension with the shoulder in 120° of abduction produced the greatest excitation of all trapezius regions.

As part of the clinical assessment of the musculoskeletal system, manual muscle tests (MMTs) are used to assess the amount of force a given muscle can produce.¹ For the trapezius muscle, this evaluation is complex, as we often describe the trapezius muscle as comprising 3 regions (upper trapezius region [UTR], middle trapezius region [MTR], and lower trapezius region [LTR]) with distinct mechanical actions. The UTR assists with clavicular elevation and upward rotation of the scapula, the MTR contributes to scapular stabilization and retraction, and the LTR contributes to inferior rotation and depression of the scapula.² Therefore, the trapezius muscle is assessed in multiple positions to evaluate each region separately and selectively recruit associated muscle fibers.¹

The trapezius muscle is often implicated in scapulothoracic and acromioclavicular joint dysfunction because of its role in scapular movement.³ Altered muscle activity of the

UTR, MTR, and LTR has been identified in people with shoulder injuries,^{4–7} most notably increased UTR excitation and decreased LTR excitation. Alterations in trapezius muscle excitation are associated with altered scapular kinematics, which are thought to increase the risk of shoulder injury.⁸ It is assumed that overactive or underactive regions of the trapezius muscle must be addressed using targeted interventions to alter the activity of these underactive or overactive regions, thereby restoring normal scapular kinematics to reduce the risk of injury.^{3,7,9} For example, if we perform an MMT for the LTR when assessing scapulothoracic joint dysfunction, acromioclavicular joint dysfunction, or both, we may conclude that the LTR is underactive or weak if the patient does not produce sufficient force. Conversely, if the patient produces sufficient force, we may assume that the LTR has adequate strength and activity to support proper scapular stabilization

and movement. In turn, we use this information in the clinical interpretation and formulation of the patient's treatment plan.

Manual muscle tests are routinely used to assess the scapula, shoulder, and neck.¹⁰ Three tests are conducted when using the Kendall MMTs: one to assess the neuromuscular function of each region.¹¹ During an MMT, we place the patient in a specific position intended to facilitate maximal isometric activation of a muscle or muscle region, with the instruction to meet the manual force applied for approximately 5 seconds. Our subjective evaluations of the force a patient produces in response to an MMT are then used to assess the neuromuscular function of the muscle, including its strength and fatigue resistance. However, despite their routine use, to our knowledge, no researchers have determined if MMTs for the trapezius preferentially recruit these regions of the trapezius. As such, the purpose of our study was to assess excitation of the UTR, MTR, and LTR individually in healthy participants during adapted MMTs. We adapted the MMTs into maximal voluntary isometric contractions (MVICs) for the upper trapezius (UT-MVIC), middle trapezius (MT-MVIC), and lower trapezius (LT-MVIC). Each MVIC was based on a Kendall MMT intended to target the corresponding region of the trapezius. Excitation was evaluated using high-density surface electromyography (HD-sEMG). We hypothesized that, for each region of the trapezius, the corresponding MVIC would produce the highest excitation relative to the other MVICs. For example, the UT-MVIC would produce greater excitation in the UTR than would the MT-MVIC and LT-MVIC.

METHODS

Participants

We recruited a convenience sample of 20 participants (10 men, 9 women; age = 23.9 ± 1.7 years, height = 171.4 ± 9.6 cm, mass = 75.7 ± 11.6 kg) from the general and student population at the University of Manitoba. Volunteers were invited to participate if they were between the ages of 18 and 40 years and self-reported right-hand dominance. Exclusion criteria are presented in Table 1. Participants provided informed consent, and the study was approved by the Education Nursing Research Ethics Board at the University of Manitoba and conformed to the Declaration of Helsinki. All data were collected in a research laboratory.

Protocol

Participants came to our laboratory for familiarization and experimental sessions, with a minimum of 48 hours between sessions. They completed 3 repetitions each, with a 2-minute rest interval, for 3 MVICs adapted from Kendall MMTs.¹¹ We randomized the order of the MVICs for each participant before data collection. As part of another project, participants performed 5 submaximal contractions at 30% or 60% MVIC, with a 1-minute rest between repetitions and a 5-minute rest between MVIC blocks to reduce fatigue (Figure 1).

Familiarization Session. Based on the randomization order (Research Randomizer version 4.0; Social Psychology Network), participants began the familiarization

Table 1. Exclusion Criteria

Any previous injuries (eg, broken collar bone, whiplash, shoulder dislocation) or orthopaedic disorders (eg, thoracic outlet syndrome, nerve root compression in the neck) in the right shoulder, upper back, or neck that affected self-reported normal function
Any neurologic (eg, epilepsy, multiple sclerosis, Parkinson disease) or musculoskeletal disorder (eg, muscular dystrophy, myasthenia gravis)
Currently experiencing pain in the shoulder, upper back, or neck

session positioned to perform an MVIC for the UTR (UT-MVIC), MTR (MT-MVIC), or LTR (LT-MVIC). For the UT-MVIC, participants sat upright on a custom apparatus and used their right hand to hold an adjustable handle, which was affixed to a force transducer (model PY6-1000; BERTEC Corp) and amplifier (model AM6501; BERTEC Corp). The force transducer has an accuracy of

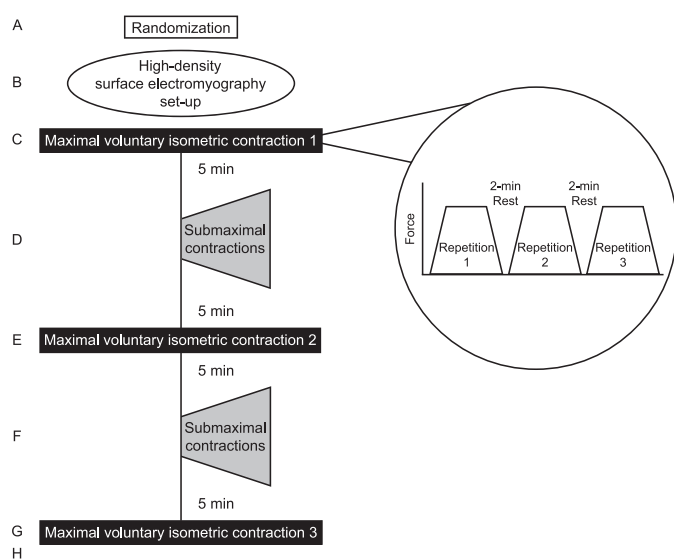


Figure 1. Experimental session overview. A, Twenty participants were screened for eligibility and then performed a randomized order of maximal voluntary isometric contractions (MVICs) for the upper (UT-MVIC), middle (MT-MVIC), and lower (LT-MVIC) regions of the trapezius. B, We applied 32-electrode, high-density surface electromyography (HD-sEMG) grids to the participant's right upper, middle, and lower trapezius regions. C, Participants performed the first MVIC (UT-MVIC, MT-MVIC, or LT-MVIC). For each MVIC, participants held a handle in their right hand that was attached to a force transducer. They were instructed to contract as hard as possible for 5 seconds. This was repeated 3 times, with a 2-minute break between repetitions. After the 3 repetitions, participants rested for 5 minutes. D, They then performed 1 set of 5 submaximal contractions for the same region as their first MVIC at 30% or 60% as part of another project. Participants rested for 1 minute between submaximal contractions. After the set, they rested for 5 minutes. E, Participants completed 3 repetitions of their second MVIC, with a 2-minute rest between repetitions. After the 3 repetitions, participants rested for 5 minutes. F, They performed 1 set of 5 submaximal contractions for the same region as their second MVIC at 30% or 60% as part of another project. Participants rested for 1 minute between submaximal contractions. After the set of submaximal contractions, they rested for 5 minutes. G, Participants completed 3 repetitions of their final MVIC, with a 2-minute rest between repetitions. After the 3 repetitions, they rested for 5 minutes. H, Participants then performed 1 set of 5 submaximal contractions for the same region as their third MVIC at 30% or 60% as part of another project. They rested for 1 minute between submaximal contractions. After the 2 sets of submaximal contractions, participants rested for 5 minutes.

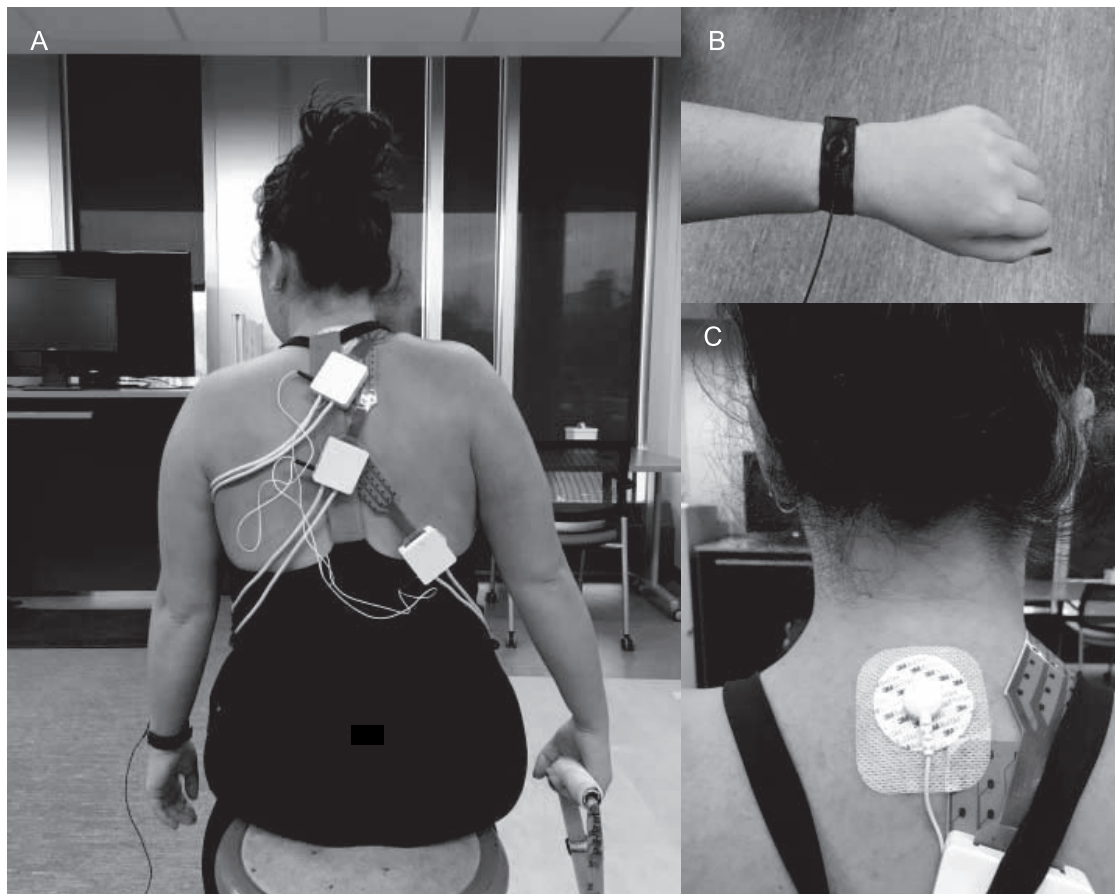


Figure 2. Upper trapezius maximal voluntary isometric contraction. **A**, Participants sat upright on a stool with their elbow fully extended, grasping an adjustable handle. We instructed them to isometrically pull the handle toward their ear, with their head rotated away from their shoulder toward a computer screen. **B**, Ground electrode at the wrist. **C**, Reference electrode at C7.

99.5% and a sensitivity of 2 mV/N.¹² Handle length was adjusted for participants so they could hold the handle directly at their side with the elbow fully extended and while seated with a neutral spinal position. We instructed participants to isometrically pull the handle toward their ear, with their head rotated away from their shoulder toward a computer screen (Figure 2).

For the MT-MVIC, participants were positioned prone on an examination table. They held the handle in their right hand at 90° to the torso (verified using a manual goniometer) in abduction and external rotation, with the handle positioned directly over the force transducer. Handle length was adjusted so that the participant's right arm was directly at his or her side in the frontal plane with minimal flexion or extension while performing the MVIC. We instructed participants to isometrically extend the shoulder and retract the scapula by attempting to pull the handle away from the floor (Figure 3).

The LT-MVIC was performed in a similar fashion to the MT-MVIC; however, participants held the handle over the force transducer with the shoulder in 120° of glenohumeral abduction (Figure 4).

For all MVICs, we asked participants to pull the handle as hard as possible, to focus on using the intended muscle for each MVIC, and to try to refrain from using other muscle groups (eg, quadriceps to push through the floor). They performed three 5-second contractions for each MVIC, during which we provided oral encouragement.

The MVICs were separated by 2 minutes of rest, and we provided participants with a 5-second countdown before initializing each MVIC. After completing the first set of MVICs (UT-MVIC, MT-MVIC, or LT-MVIC), participants were given 5 minutes of rest before performing submaximal isometric contractions in the same position, followed by another 5-minute rest. Completion of the MVICs and submaximal contractions for the second and third movements followed, with procedures mirroring those of the first (Figure 1).

Experimental Session. The experimental session mirrored the familiarization session, with the addition of the collection of HD-sEMG measurements from the UTR, MTR, and LTR during the MVICs.

High-Density Surface Electromyography

Before electrode application, the participant's skin was shaved (if necessary), scrubbed using abrasive paste (Nuprep; Weaver and Company), and wiped clean using 70% isopropyl alcohol. To achieve optimal electromyography (EMG) signal quality, we identified the innervation zone for the UTR, MTR, and LTR fibers as described by Barbero et al.¹³ Next, we marked the innervation zone on the participant's skin and placed an electrode grid medial to each zone.

We detected HD-sEMG signals from the UTR, MTR, and LTR using three 2-dimensional grids of 32 gold-coated

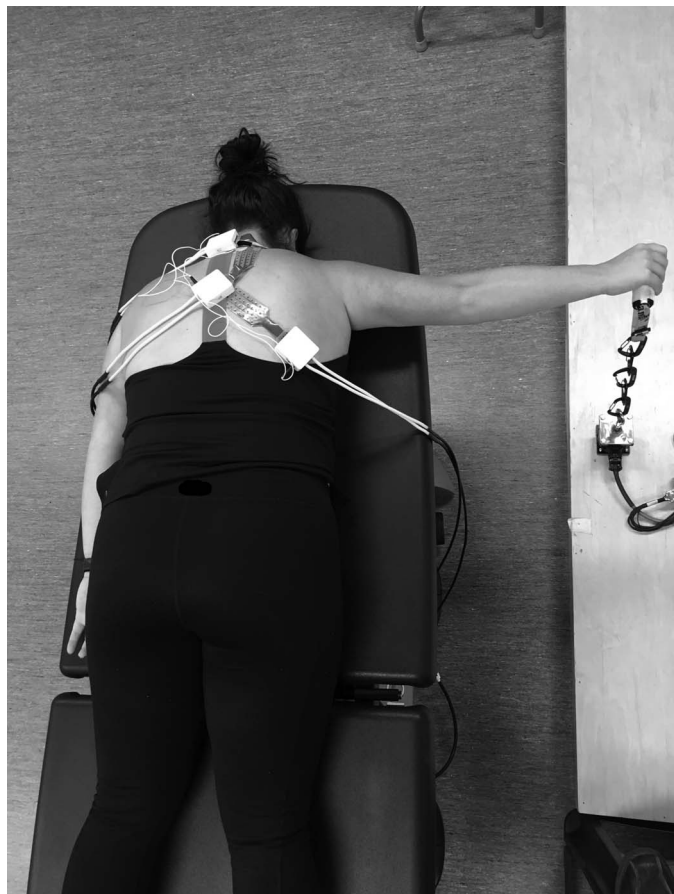


Figure 3. Middle trapezius maximal voluntary isometric contraction. Participants were positioned prone on an examination table. They held the handle at 90° to the torso in abduction and external rotation, with the handle directly over the force transducer. We instructed them to isometrically extend the shoulder and retract the scapula by attempting to pull the handle away from the floor.



Figure 4. Lower trapezius maximal voluntary isometric contraction. Participants were positioned prone on an examination table. They held the handle at 120° to the torso in abduction and external rotation, with the handle directly over the force transducer. We instructed them to isometrically extend the shoulder and retract the scapula by attempting to pull the handle away from the floor.

electrodes (model GR10MM0804; OT Bioelettronica) with an interelectrode distance of 10 mm arranged in 4 columns and 8 rows. We placed ground electrodes around the left wrist and at the C7 spinous process. Electrode grids were positioned with rows of electrodes aligned approximately parallel with muscle-fiber direction and adhered to the skin using disposable adhesive foam matrices (model KI-TAD4x8NM6; OT Bioelettronica) that were filled with conductive paste (Ten20; Weaver and Company) and secured to the skin using tape (Hypafix; BSN Medical).

Signal Analysis and Processing

Force transducer and HD-sEMG data were collected during each MVIC using a 400-channel bioelectrical signal amplifier with a common mode rejection ratio > 95 dB (Quattrocento; OT Bioelettronica). Raw data were measured in volts and collected and analyzed using OT BioLab+ (OT Bioelettronica) software. The HD-sEMG and force-transducer signals were collected at 2048 Hz and bandpass filtered at 10 to 500 Hz with a bandwidth of -3 dB. We applied a scaling factor of 1000 based on manufacturer guidelines to the force transducer signal to convert to newtons, with a 10-mV offset. We further high-pass filtered HD-sEMG signals using a 50-Hz, second-order Butterworth filter to reduce electrocardiographic noise and

DC offset.^{14,15} All channels were visually inspected to identify poor signals (ie, poor contact, movement artifact), and identified channels were removed. We then replaced these channels by averaging the values of the electrodes in the corresponding row up to a maximum of 4 electrodes per grid and 1 per row. After we inspected and cleaned the signal, a total of 13 participants remained for the UTR grid root mean square (RMS), while 18 participants remained for both the MTR and LTR grid RMS; 1 participant did not complete the experimental session. We then calculated the differential of adjacent monopolar signals within each row, producing 3 differential signals for each row. Differential signals within the grid were averaged, and RMS was calculated to provide an average RMS for each electrode grid, representing excitation of each region of the trapezius.

Statistical Analysis

We exported data as a .csv file in 0.5-second epochs for each 5-second MVIC. The last 6 epochs (3 seconds) for each grid (UTR, MTR, LTR) were averaged and used for data analysis. We then transferred exported data into a custom Excel (version 2016; Microsoft Corp) spreadsheet and then into SPSS (version 25; IBM Corp) for inferential statistical analysis. Inferential statistical analyses were performed on participants' peak force-producing trial for

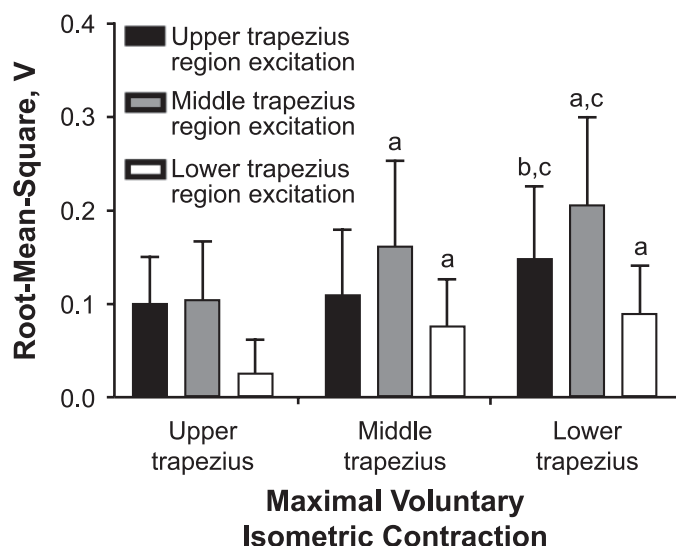


Figure 5. One-way analysis-of-variance results for upper trapezius, middle trapezius, and lower trapezius region excitation during maximal voluntary isometric contractions (MVICs). ^a Different from UT-MVIC ($P < .001$). ^b Different from MT-MVIC ($P < .017$). ^c Different from LT-MVIC ($P < .001$).

each MVIC. We conducted three 1-way analyses of variance (ANOVAs) for repeated measures to determine the effects of each modified muscle test on muscle excitation of the UTR, MTR, and LTR. A Bonferroni correction ($P = .05/3$ comparisons) was applied for each ANOVA, resulting in $\alpha = 0.017$.

Outliers were determined using boxplot inspection. In SPSS, values > 1.5 box lengths are considered outliers. For the MT grid RMS, participant 5 was an outlier for both the UT-MVIC and LT-MVIC. For the LT grid RMS, participants 5, 8, and 15 were outliers for the UT-MVIC. Statistical analyses were conducted with and without outliers. Outliers did not affect the outcomes of the statistical tests. As such, they were left in the analysis (Supplemental Table).

We assessed normality using the Shapiro-Wilk test ($P > .05$). Data were normally distributed for all dependent variables except MTR grid RMS for the UT-MVIC and LTR grid RMS for the UT-MVIC. For these variables, we also performed the nonparametric Friedman test to determine if normality affected the results (Supplemental Table). We assessed sphericity using the Mauchly test ($P > .05$). If the assumption of sphericity was not met, a Greenhouse-Geisser correction was applied. The 1-way ANOVA and Friedman test agreed; therefore, given the robustness of the 1-way ANOVA to violations of the assumption of normality, we only report the results of the 1-way ANOVAs. We calculated post hoc Bonferroni pairwise comparisons for ANOVAs that were significant. Effect sizes were measured using partial η^2 , with 0.01 indicating a

small effect size; 0.06, a medium effect size; and 0.14, a large effect size.¹⁶

We determined the intraclass correlation coefficients (ICCs) for grid RMS during each MVIC to evaluate the reliability of our excitation measure across the 3 trials. A 2-way mixed-model absolute agreement was used for analysis of each ICC. Those ICCs < 0.5 were considered *poor*; from 0.5 to 0.75, *moderate*; from 0.75 to 0.9, *good*; and > 0.9 , *excellent reliability*.¹⁷

RESULTS

Descriptive statistics for excitation are presented in Table 2. Mean $\pm$ SD peak force production for the UT-MVIC was 435.5 ± 172.0 N, MT-MVIC was 53.7 ± 25.2 N, and LT-MVIC was 50.3 ± 27.7 N.

Excitation of the UTR

The Mauchly test indicated that the assumption of sphericity was not met ($\chi^2 = 14.085$, $P = .001$). As such, a Greenhouse-Geisser correction was applied. We observed a difference across MVICs, with a large effect size for UTR excitation ($F_{1,161,13.937} = 10.567$, $P = .005$, partial $\eta^2 = 0.468$). Post hoc analysis indicated an increase in UTR excitation during the LT-MVIC compared with the UT-MVIC (0.048 V [95% CI = 0.009, 0.088 V]; $P = .016$) and MT-MVIC (0.03 V [95% CI = 0.025, 0.053 V]; $P < .001$; Figure 5).

Excitation of the MTR

We observed a difference across MVICs, with a large effect size for MTR excitation ($F_{2,34} = 44.001$, $P < .001$, partial $\eta^2 = 0.721$). Post hoc analysis demonstrated an increase in MTR excitation during the MT-MVIC (0.057 V [95% CI = 0.024, 0.090 V]; $P = .001$) and LT-MVIC (0.102 V [95% CI = 0.072, 0.132 V]; $P < .001$) compared with the UT-MVIC. We also noted an increase in MTR excitation during the LT-MVIC compared with the MT-MVIC (0.045 V [95% CI = 0.022, 0.068 V]; $P < .001$; Figure 5).

Excitation of the LTR

A difference was evident across MVICs, with a large effect size for LTR excitation ($F_{2,34} = 19.370$, $P < .001$, partial $\eta^2 = 0.521$). Post hoc analysis indicated an increase in LTR excitation during the MT-MVIC (0.051 V [95% CI = 0.023, 0.078 V]; $P < .001$) and LT-MVIC (0.064 V [95% CI = 0.038, 0.089 V]; $P < .001$) compared with the UT-MVIC (Figure 5).

Intraclass Correlation Coefficient

The ICC values are presented in Table 3. Coefficient values for both the MT-MVIC and LT-MVIC were > 0.9 .

Table 2. Descriptive Statistics for the Upper, Middle, and Lower Trapezius Excitation During Maximal Voluntary Isometric Contractions

Trapezius Region Maximal Voluntary Isometric Contraction	Trapezius Region Root Mean Square, Mean $\pm$ SD, V		
	Upper (n = 13)	Middle (n = 18)	Lower (n = 18)
Upper	0.1013 $\pm$ 0.0500	0.1049 $\pm$ 0.0625	0.0269 $\pm$ 0.0359
Middle	0.1105 $\pm$ 0.0691	0.1622 $\pm$ 0.0920	0.0777 $\pm$ 0.0494
Lower	0.1494 $\pm$ 0.0774	0.2072 $\pm$ 0.0936	0.0904 $\pm$ 0.0514

Table 3. Intraclass Correlation Coefficients Across Trials for Excitation of the Upper, Middle, and Lower Trapezius During Maximal Voluntary Isometric Contractions

Trapezius Region Maximal Voluntary Isometric Contraction	Trapezius Region Root Mean Square Intraclass Correlation Coefficient		
	Upper	Middle	Lower
Upper	0.969	0.984	0.622
Middle	0.989	0.991	0.992
Lower	0.986	0.962	0.989

DISCUSSION

The purpose of our study was to assess the excitation of the UTR, MTR, and LTR individually during 3 MVICs, which were adapted from the Kendall MMTs and designed to target the respective regions of the trapezius. The MMTs are a common tool used by clinicians to assess the strength of individual muscles through specific MVICs.¹⁰ Generally, a patient is placed in the appropriate position and instructed to meet the manual force applied by the clinician for approximately 5 seconds. These testing procedures allow us to subjectively evaluate the patient's response specific to the tested muscle, including strength and fatigue resistance. To assess the scapula, shoulder, or neck, individual MMTs are performed for the 3 regions of the trapezius; each is thought to preferentially test and recruit the respective region of the trapezius.¹¹ We hypothesized that, for each region of muscle fibers, the MVIC that was designed to target it would elicit the greatest excitation relative to the 2 other MVICs. However, based on our electromyographic findings, the UT-MVIC and MT-MVIC did not maximally recruit the targeted regions of the trapezius more than the other tests did. In agreement with our hypothesis, the LT-MVIC was the only MVIC that produced greater excitation for the targeted region (ie, LTR) compared with the other MVICs. Interestingly, the LT-MVIC produced the greatest excitation in all regions of the trapezius.

Although widely used clinically and often described as producing maximal contractions,¹⁸ trapezius MMTs for the UT and MT may not produce maximal excitation of the intended muscles in the shoulder complex. Using the Kendall MMTs to normalize their EMG data, Ekstrom et al¹⁸ noted that participants who performed a dynamic shrug without neck rotation produced a normalized mean of $119\% \pm 23\%$ MVIC in the UTR. In a follow-up study, Ekstrom et al¹⁹ found that, indeed, an MMT involving scapular elevation with neck extension and rotation away from the shoulder produced a mean of only $82\% \pm 16\%$ MVIC in the UTR when normalized to the maximal EMG values produced by the other MMTs studied. In contrast, shoulder flexion- and shoulder abduction-based MMTs produced means of $83\% \pm 13\%$ MVIC and $92\% \pm 9\%$ MVIC in the UTR, respectively.

In more recent work, Cibulka et al²⁰ investigated the potential for assessing the trapezius as a whole using 1 whole-muscle MMT. The EMG data obtained during the Kendall UTR MMT (scapular elevation with side flexion and rotation away from the shoulder) were used to normalize the EMG signals. Their proposed whole-muscle MMT, involving resisted shoulder abduction and internal rotation, produced a mean of $160.84\% \pm 58.78\%$ MVIC in the UTR. Similar to the reports of Ekstrom et al,^{18,19} these

findings suggested that shoulder abduction produced substantial excitation in the UTR. In addition, these data implied that the Kendall UTR MMTs did not produce a maximal contraction in the UTR, as the whole trapezius MMT produced approximately 61% more EMG excitation than did the Kendall UTR MMT. These results are consistent with ours, as the MVIC adapted from the Kendall UTR MMT produced the least amount of excitation in the UTR across the 3 MVICs examined (Figure 5).

The clinical utility of MMTs is inherent to their ability to test strength and muscle function of specific muscles. Based on the aforementioned studies, if an MMT is unable to elicit maximal contractions in the intended muscles of healthy individuals, then it will provide limited clinical information regarding specific muscle function or strength. Rather, interpreting the amount of force produced by an MMT that did not produce maximal contractions within a given muscle would only supply general information regarding muscle functioning of the synergists and coactivators for the movement. The interpretation of MMTs as producing maximal contractions in a specific muscle, therefore, may misdirect the clinical interpretation.

We observed greater UTR excitation during the LT-MVIC than the UT-MVIC and MT-MVIC. Despite applying resistance in the sagittal plane (ie, making the shoulder resist extension while positioned in 120° of abduction), the LT-MVIC was able to produce the greatest excitation of the 3 MVICs. Shoulder abduction to 90° with side flexion of the neck and rotation to the opposite side with resistance applied toward the ground produced the greatest UTR excitation in the Ekstrom et al¹⁹ study. In addition, as previously mentioned, Cibulka et al²⁰ found that resisted shoulder abduction produced substantial UTR excitation in comparison with the Kendall UTR MMT. Collectively, it would appear that placing the shoulder in abduction, regardless of whether resistance is applied in the plane in which the muscle acts, helps to elicit greater excitation of the UTR during MMTs or MVICs compared with MMTs or MVICs involving scapular elevation exclusively.

Whereas the inability of the UT-MVIC to produce the greatest excitation in the UTR was unexpected, a few explanations, including the recruitment of secondary and accessory muscles, are probable. Hintermeister et al²¹ recorded EMG for multiple muscles in the shoulder complex while participants performed 7 common shoulder rehabilitative exercises using resistance bands, including a shrug, which was similar to our UT-MVIC. During the elevation phase of the shrug exercise, the shoulder muscles other than the trapezius were substantially excited, with the subscapularis being the most excited muscle based on the MVIC %. In turn, this may suggest that force from secondary muscles contributed substantially to the UT-MVIC in our investigation. This is speculative, however, as we did not record EMG from muscles other than the trapezius.

In addition to secondary muscle contributions, individual variability may also have affected our results. Individual differences in motor-recruitment strategies have been demonstrated in the trapezius muscle.²² Furthermore, multiple authors^{23,24} determined that the absolute maximum excitation may not be generated for each muscle in every

participant when examining the trapezius and shoulder muscles. As such, depending on the MMT or MVIC, it is possible that the level of excitation in each region of the trapezius varies significantly based on the large SDs present (Table 2). Indeed, some participants did produce a higher level of UTR excitation from the UT-MVIC; however, this was not the general trend of our data.

Interestingly, the LT-MVIC descriptively produced the greatest excitation of all regions. Although we expected that the LT-MVIC would produce the greatest excitation in the LTR, the greater excitation produced in the MTR from the LT-MVIC compared with the MT-MVIC may pose concerns for MTR MMT validity. Similar to us, Ekstrom et al¹⁸ found that raising the shoulder overhead in line with the LTR fibers, a movement indicated for preferentially recruiting the LTR,¹¹ produced the greatest MTR excitation ($101\% \pm 32\%$ MVIC). Yet previous researchers¹⁸ proposed that horizontal abduction with external rotation, rather than other MMTs, provided the greatest MTR excitation. As such, across these studies, it is unclear which MMT is more appropriate for assessing the MTR fibers. Alternatively, given that the UTR is involved in scapular elevation,² our results may suggest that participants were using abduction to resist the movement rather than the intended extension. Our experimental setup did not restrict motion in the frontal plane, so we do not know if this was the case.

Although the UT-MVIC did not produce the greatest excitation in the UT, it may still serve as a useful test for EMG normalization. Across the 3 trials for each MVIC, the ICC for the peak excitation of the corresponding regions was > 0.9 , indicating very high reliability at least within sessions. In turn, from a reliability perspective, our UT-MVICs, MT-MVICs, and LT-MVIC data could be useful for normalizing EMG data.

Given that participants pulled against a handle rather than pushing against a clinician's force, the MVICs used are modified from their original form as described by Kendall et al.¹¹ Thus, our findings and interpretations of these data are limited to the MVICs used in our study. Other MVICs or MMTs may result in greater peak muscle excitation. Furthermore, MMTs are normally collected as part of the assessment of injured shoulders, and our population was healthy. Therefore, it is possible that these MVICs may produce different results in the intended population.

In our work, 1 researcher performed the MVIC positioning and supplied instructions to participants for consistency. Because participants were applying maximal force to a fixed object through the pull of the handle, the amount of force they would have had to push against did not differ. They were instructed to pull the handle as hard as they possibly could, but they were not strapped to the table or apparatus. It is possible that participants were not exerting maximal effort or were using additional muscles when performing the MVICs. Additional "noise" may have been added to the data by the submaximal contractions that participants completed between MVICs, although they were permitted substantial rest to reduce the potential effect of fatigue. Furthermore, MVIC and submaximal contraction intensity were randomized and counterbalanced, which should have minimized any effect on the data.

CONCLUSIONS

We assessed excitation of the UTR, MTR, and LTR individually during 3 MVICs. Only the LT-MVIC demonstrated the greatest peak excitation in the targeted region. Therefore, the adapted MVICs for the UTR and MTR may not efficiently excite the corresponding regions of the trapezius. Given our small sample size and the likelihood of large interindividual variability, more research is needed to determine if these results are stable in larger populations with injuries. In the meantime, clinicians should consider that multiple assessments of strength and force may be needed to evaluate neurologic function in the individual regions of the trapezius, and these test results should be considered holistically in the context of the patient. Indeed, it is possible that an MVIC for the trapezius involving abduction and resisted extension may provide information on trapezius function as a whole; however, further exploration is needed.

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